Samarth Kumar Samal

Data Scientist | Healthcare Analytics | Software Engineering
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PROFESSIONAL SUMMARY

M.S Data Science student at the University of Minnesota with 1 year of software engineering experience at Dell Technologies and current research experience in healthcare analytics. Experienced in Python, SQL, machine learning, electronic health record analysis, signal processing and reproducible data workflows. Background includes solving technical problems, analyzing complex datasets, applying machine learning and statistical methods, and collaborating with research and cross-functional engineering teams.

EDUCATION

University of Minnesota, Twin Cities — **2025–2027**
Master of Science in Data Science; Minor in Computer Science; GPA: 3.61/4.00 — *Minneapolis, Minnesota*

Odisha University of Technology and Research — **2018–2022**
Bachelor of Technology in Electrical Engineering; GPA: 3.83/4.00 — *Bhubaneswar, India*

TECHNICAL SKILLS

Programming & Databases: Python, SQL, R, C++
Machine Learning & Statistics: scikit-learn, Classification, Clustering, Feature Engineering, Cross-Validation, Model Evaluation, Statistical Analysis
Data Analysis: Pandas, NumPy, SciPy, Data Preprocessing, Missing Data Handling
Healthcare Data & Clinical Analytics: eICU, MIMIC-III, Electronic Health Record Analysis, Clinical Data Preprocessing
Signal Processing: IMU Data, Time-Domain Analysis, Frequency-Domain Analysis, Power Spectral Density, Spectrogram Analysis
Visualization & Applications: Matplotlib, Seaborn, Plotly, Streamlit
Productivity & Collaboration: Microsoft Excel, Microsoft Word, Microsoft PowerPoint, Google Sheets, Google Docs, Google Slides, Google Drive
Tools & Platforms: Git, GitHub, Docker, Google Cloud Platform (GCP), Linux, JupyterLab, Visual Studio Code, Jira

RESEARCH EXPERIENCE

Graduate Researcher — **May 2026–Present**
Neuromodulation Center, Department of Neurology — *Minneapolis, Minnesota*
Principal Investigator: Prof. Biswaranjan Mohanty

- Analyze raw biomedical sensor time-series data using statistical and exploratory methods to identify signal patterns, assess data quality, and guide preprocessing decisions.
- Develop reproducible data-processing workflows for signal cleaning, transformation, resampling, segmentation, and feature engineering to prepare sensor data for machine-learning classification and modeling tasks.
- Apply statistical and signal-processing methods to extract temporal features and improve the quality of model-ready datasets for downstream analysis.
- Train and tune seven machine-learning models, including Random Forest and Extra Trees, achieving a Macro-F1 score of 0.60.

Capstone Researcher — **May 2026–Present**
Department of Surgery, Division of Computational Health Sciences — *Minneapolis, Minnesota*
Principal Investigator: Prof. Jocelyn Chi

- Build a reproducible healthcare data-science pipeline to transform complex ICU electronic health records into structured, model-ready datasets for patient-level analysis.
- Perform data-quality assessment, missing-value analysis, feature engineering, and normalization to improve data reliability and support consistent downstream modeling.
- Apply exploratory and statistical analysis to uncover patient-level patterns, evaluate feature behavior, and guide data-preprocessing and modeling decisions.
- Develop and compare unsupervised machine-learning models using clustering quality, stability, and interpretability metrics to identify meaningful patient subtypes.
- Evaluate discovered patient groups against clinical outcomes and severity measures to determine whether data-driven patterns provide meaningful healthcare insights.

PROFESSIONAL EXPERIENCE

Software Engineer I — **July 2022–August 2023**
Dell Technologies — *Bengaluru, India*

- Worked in the Dell Update Package (DUP) and USBNIC teams, supporting 5 component areas: iDRAC, USBNIC, PSU, CPLD, and device drivers across Windows, WinPE, and Linux environments.
- Resolved more than 50 Jira issues and contributed to over 100 technical investigations through initial analysis, log review, source-code debugging, issue escalation, testing, and fix validation.
- Contributed to the modernization of DUP libraries by supporting development, testing, and validation of Dell Update Package workflows and component updates.
- Collaborated with the U.S.-based engineering team on the OSM module to analyze issues, validate functionality, and coordinate technical requirements across development and testing teams.
- Set up and supported more than 50 virtual machines and 10 physical servers with the Cloud Server team to meet development, testing, operating-system, and platform requirements.

TECHNICAL PROJECTS

Temporal Clinical Reasoning for Medical Coding — **May 2026**
GitHub Repository | **Live Application** — *BioClinicalBERT, Longformer, BiLSTM, MIMIC-III*

- Developed an end-to-end clinical NLP pipeline using BioClinicalBERT, Longformer, and BiLSTM models to predict diagnosis and procedure codes from longitudinal electronic health record notes.
- Built structured patient-level datasets from MIMIC-III by cleaning and organizing clinical notes, preserving temporal relationships, mapping medical codes, and preparing data for model training and evaluation.
- Trained and compared multiple deep-learning architectures using Micro-F1, Macro-F1, and Precision@5, with Longformer achieving a 0.5925 joint score and the temporal model reaching 0.6096 CPT Micro-F1.
- Analyzed temporal patterns across clinical documentation to evaluate whether longitudinal patient information improves context-aware medical code prediction compared with independent note-level modeling.

COVID-19 Forecasting Web Application — **December 2025**
GitHub Repository | **Live Application** — *Python, Prophet, Plotly*

- Built and deployed an interactive data-analytics application using Python and Streamlit, processing 161,568 records across 198 countries and 816 reporting dates.
- Developed reusable data-cleaning and analysis workflows to transform large time-series datasets into structured insights, trends, and country-level performance metrics.
- Implemented configurable time-series forecasting using Prophet with weekly and yearly seasonality and 95% prediction intervals to model future country-level trends.
- Designed interactive dashboards, maps, KPI summaries, comparisons, rankings, and downloadable reports using Plotly, Matplotlib, and Streamlit to communicate analytical results clearly.